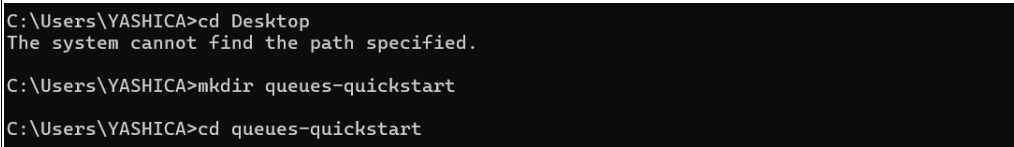


EXPERIMENT 5

Name: Shraddha Desai		Roll no: 18
Div: A		Batch: A
Title	To study and Implement Storage as a Service using Azure Storage- Queues and Tables .	
Pre requisite	NIST model of Cloud Computing	
Mapping with CO	CSL605.5	
Objective	To explore Queues and Tables on Microsoft Azure Storage.	
Outcome	To assess and compare commercially available cloud services, recommending the most suitable one for a given application.	
Instructions	Follow following links to perform this experiment for Queue Data storage: 1. https://learn.microsoft.com/en-us/azure/storage/queues/storage-queues-introduction 2. https://learn.microsoft.com/en-us/azure/storage/queues/storage-quickstart-queues-python?tabs=passwordless%2Croles-azure-portal%2Cenvironment-variable-windows%2Csign-in-azure-cli	
Deliverables	Submit stepwise screenshots for creating python app and incorporating queue data storage service into it.	
Output	Procedure Step 1: Create Project Directory Open command prompt and execute: mkdir queues-quickstart cd queues-quickstart  <pre>C:\Users\YASHICA>cd Desktop The system cannot find the path specified. C:\Users\YASHICA>mkdir queues-quickstart C:\Users\YASHICA>cd queues-quickstart</pre>	
	Step 2: Install Required Libraries pip install azure-storage-queue azure-identity	

```
c:\Users\YASHICA\queues-quickstart>pip install azure-storage-queue azure-identity
Defaulting to user installation because normal site-packages is not writeable
Collecting azure-storage-queue
  Downloading azure_storage_queue-12.15.0-py3-none-any.whl.metadata (37 kB)
Collecting azure-identity
  Downloading azure_identity-1.25.3-py3-none-any.whl.metadata (91 kB)
Collecting azure-core>=1.30.0 (from azure-storage-queue)
  Downloading azure_core-1.39.0-py3-none-any.whl.metadata (48 kB)
Collecting cryptography>=2.1.4 (from azure-storage-queue)
  Downloading cryptography-46.0.6-cp311-abi3-win_amd64.whl.metadata (5.7 kB)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\users\yashica\appdata\roaming\python\python313\site-packages (from
0)
Collecting isodate>=0.6.1 (from azure-storage-queue)
  Downloading isodate-0.7.2-py3-none-any.whl.metadata (11 kB)
Collecting msal>=1.35.1 (from azure-identity)
  Downloading msal-1.35.1-py3-none-any.whl.metadata (11 kB)
Collecting msal-extensions>=1.2.0 (from azure-identity)
  Downloading msal_extensions-1.3.1-py3-none-any.whl.metadata (7.8 kB)
Requirement already satisfied: requests>=2.21.0 in c:\users\yashica\appdata\roaming\python\python313\site-packages (from azure-c
ueue) (2.32.5)
Collecting cffi>=2.0.0 (from cryptography>=2.1.4->azure-storage-queue)
  Downloading cffi-2.0.0-cp313-cp313-win_amd64.whl.metadata (2.6 kB)
Collecting pycparser (from cffi>=2.0.0->cryptography>=2.1.4->azure-storage-queue)
  Downloading pycparser-3.0-py3-none-any.whl.metadata (8.2 kB)
Collecting PyJWT<3,>=1.0.0 (from PyJWT[crypto]<3,>=1.0.0->msal>=1.35.1->azure-identity)
  Downloading pyjwt-2.12.1-py3-none-any.whl.metadata (4.1 kB)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\yashica\appdata\roaming\python\python313\site-packages (from
e>=1.30.0->azure-storage-queue) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in c:\users\yashica\appdata\roaming\python\python313\site-packages (from requests>=2
azure-storage-queue) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\yashica\appdata\roaming\python\python313\site-packages (from requ
0.0->azure-storage-queue) (2.6.1)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\yashica\appdata\roaming\python\python313\site-packages (from requ
0.0->azure-storage-queue) (2025.11.12)
Downloading azure_storage_queue-12.15.0-py3-none-any.whl (187 kB)
Downloading azure_identity-1.25.3-py3-none-any.whl (192 kB)
Downloading azure_core-1.39.0-py3-none-any.whl (218 kB)
Downloading cryptography-46.0.6-cp311-abi3-win_amd64.whl (3.5 MB)
3.5/3.9 MB 1.7 MB/s 0:00:01
Downloading cffi-2.0.0-cp313-cp313-win_amd64.whl (183 kB)
```

Step 3: Create Python File

Create a file named:
queues-quickstart.py

```
import os, uuid
from azure.identity import DefaultAzureCredential
from azure.storage.queue import QueueServiceClient, QueueClient, QueueMessage

try:
    print("Azure Queue storage - Python quickstart sample")

    # Code will go here

except Exception as ex:
    print('Exception:')
    print(ex)
```

Step 4: Create Storage Account in Azure

1. Open Azure Portal
2. Click Create → Storage Account
3. Enter details:

- Resource Group: queue-rg
 - Storage Account Name: unique name
 - Region: nearest location
4. Click Review + Create → Create

Create a storage account ...

Basics Advanced Networking Data protection Encryption Tags Review + create

[View automation template](#)

Basics

Subscription	Azure for Students
Resource group	queue-rg
Location	Korea Central
Storage account name	queuestorage12345
Preferred storage type	
Performance	Standard
Replication	Read-access geo-redundant storage (RA-GRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Managed Identity for SMB	Disabled

Security

Secure transfer	Enabled
Blob anonymous access	Disabled
Allow storage account key access	Enabled

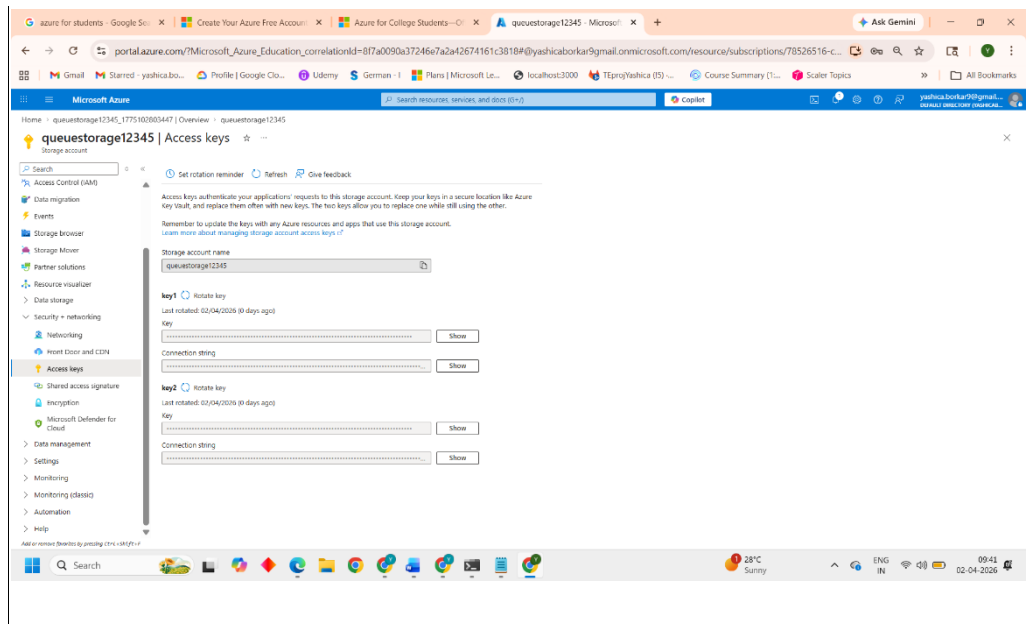
Previous

Next

Create

Step 5: Obtain Connection String

1. Open the created storage account
2. Navigate to Access keys
3. Copy the Connection string (Key1)



Step 6: Configure Environment Variable

Execute the following command:

```
setx AZURE_STORAGE_CONNECTION_STRING  
"YOUR_CONNECTION_STRING"
```

Restart the command prompt after setting the variable.

```
C:\Users\VASHICA\queues-quickstart>notepad queues-quickstart.py  
C:\Users\VASHICA\queues-quickstart>setx AZURE_STORAGE_CONNECTION_STRING "DefaultEndpointsProtocol=https;AccountName=queststorage12345;AccountKey=hbo61+VLBkf  
x2622Wl7ImacF9nE26yC12DSF+kh3uN1BkzPEVe8hJeyV66pUZfyyCLM4hsFG/L/G+ASTWY6dTg==;EndpointSuffix=core.windows.net"  
SUCCESS: Specified value was saved.  
C:\Users\VASHICA\queues-quickstart>
```

Step 7: Write Python Code

```
import os, uuid
```

```
from azure.identity import DefaultAzureCredential
```

```
from azure.storage.queue import QueueServiceClient, QueueClient,  
QueueMessage
```

```
try:
```

```
    print("Azure Queue storage - Python quickstart sample")
```

```
    # Retrieve connection string
```

```
    connect_str = os.getenv('AZURE_STORAGE_CONNECTION_STRING')
```

```
    # Create unique queue name
```

```
    queue_name = "quickstartqueues-" + str(uuid.uuid4())
```

```
    print("Creating queue: " + queue_name)
```

```
    # Create Queue Client
```

```

    queue_client = QueueClient.from_connection_string(connect_str,
queue_name)

    # Create the queue
    queue_client.create_queue()

    print("\nAdding messages to the queue...")

    queue_client.send_message("First message")
    queue_client.send_message("Second message")
    saved_message = queue_client.send_message("Third message")

except Exception as ex:
    print('Exception:')
    print(ex)

print("\nPeek at the messages in the queue...")

peeked_messages = queue_client.peak_messages(max_messages=5)

for msg in peeked_messages:
    print("Message:", msg.content)

print("\nUpdating the third message in the queue...")

queue_client.update_message(
    saved_message,
    pop_receipt=saved_message.pop_receipt,
    content="Third message has been updated"
)

print("\nPeek after update...")

peeked_messages = queue_client.peak_messages(max_messages=5)

for msg in peeked_messages:
    print("Message:", msg.content)

print("\nReceiving messages from the queue...")

messages = queue_client.receive_messages(max_messages=5)

for msg_batch in messages.by_page():
    for msg in msg_batch:
        print("Processing:", msg.content)

```

```
queue_client.delete_message(msg)
```

```
print("\nDeleting queue...")
queue_client.delete_queue()
print("Done")
```

Step 8: Run the Program

```
python queues-quickstart.py
```

Create Queue

```
PS C:\Users\YASHICA> cd queues-quickstart
PS C:\Users\YASHICA\queues-quickstart> python queues-quickstart.py
Azure Queue storage - Python quickstart sample
Creating queue: quickstartqueues-a0b143f0-2ce6-4696-82ca-15e66a2c971f
PS C:\Users\YASHICA\queues-quickstart> |
```

Step 9: Verify Output

- Messages are added, updated, and processed
- Queue is created and deleted successfully

Add Message to Queue

```
Creating queue: quickstartqueues-a0b143f0-2ce6-4696-82ca-15e66a2c971f
PS C:\Users\YASHICA\queues-quickstart> python queues-quickstart.py
Azure Queue storage - Python quickstart sample
Creating queue: quickstartqueues-9c3a2773-ba18-4acc-8e85-be3d69fe87df

Adding messages to the queue...
PS C:\Users\YASHICA\queues-quickstart> |
```

Peek Messages

```
Adding messages to the queue...
PS C:\Users\YASHICA\queues-quickstart> python queues-quickstart.py
Azure Queue storage - Python quickstart sample
Creating queue: quickstartqueues-56328416-5627-47bf-be2b-ac67bf63a476

Adding messages to the queue...

Peek at the messages in the queue...
Message: First message
Message: Second message
Message: Third message
PS C:\Users\YASHICA\queues-quickstart> |
```

Updating Message

```
PS C:\Users\YASHICA\queues-quickstart> python queues-quickstart.py
Azure Queue storage - Python quickstart sample
Creating queue: quickstartqueues-cdf723f1-1d9d-412e-a8ec-c81300409d93

Adding messages to the queue...

Peek at the messages in the queue...
Message: First message
Message: Second message
Message: Third message

Updating the third message in the queue...

Peek after update...
Message: First message
Message: Second message
Message: Third message has been updated
PS C:\Users\YASHICA\queues-quickstart> |
```

Receiving message and Deleting queue

```
Processing: Third message has been updated
PS C:\Users\YASHICA\queues-quickstart> python queues-quickstart.py
Azure Queue storage - Python quickstart sample
Creating queue: quickstartqueues-30d34f25-183e-4149-a547-20ac26f27afa

Adding messages to the queue...

Peek at the messages in the queue...
Message: First message
Message: Second message
Message: Third message

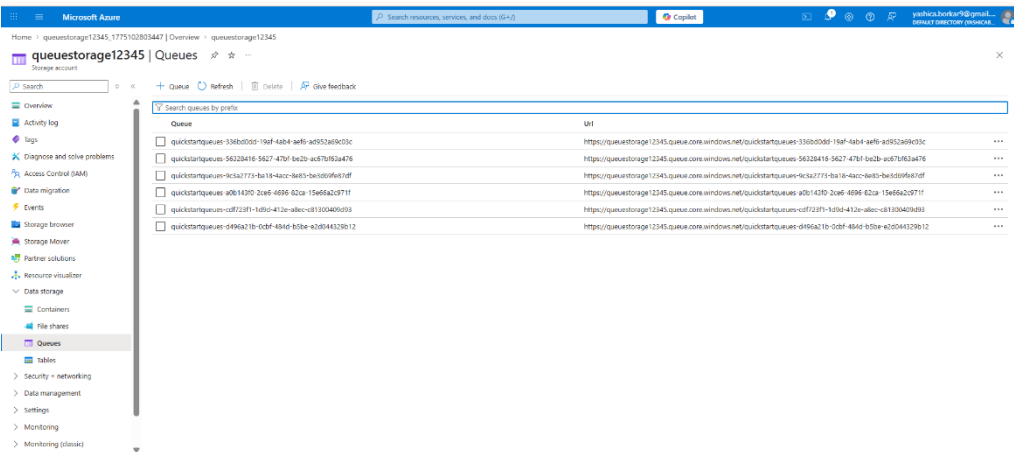
Updating the third message in the queue...

Peek after update...
Message: First message
Message: Second message
Message: Third message has been updated

Receiving messages from the queue...
Processing: First message
Processing: Second message
Processing: Third message has been updated

Deleting queue...
Done
PS C:\Users\YASHICA\queues-quickstart>
```

Step 10: Verify in Azure



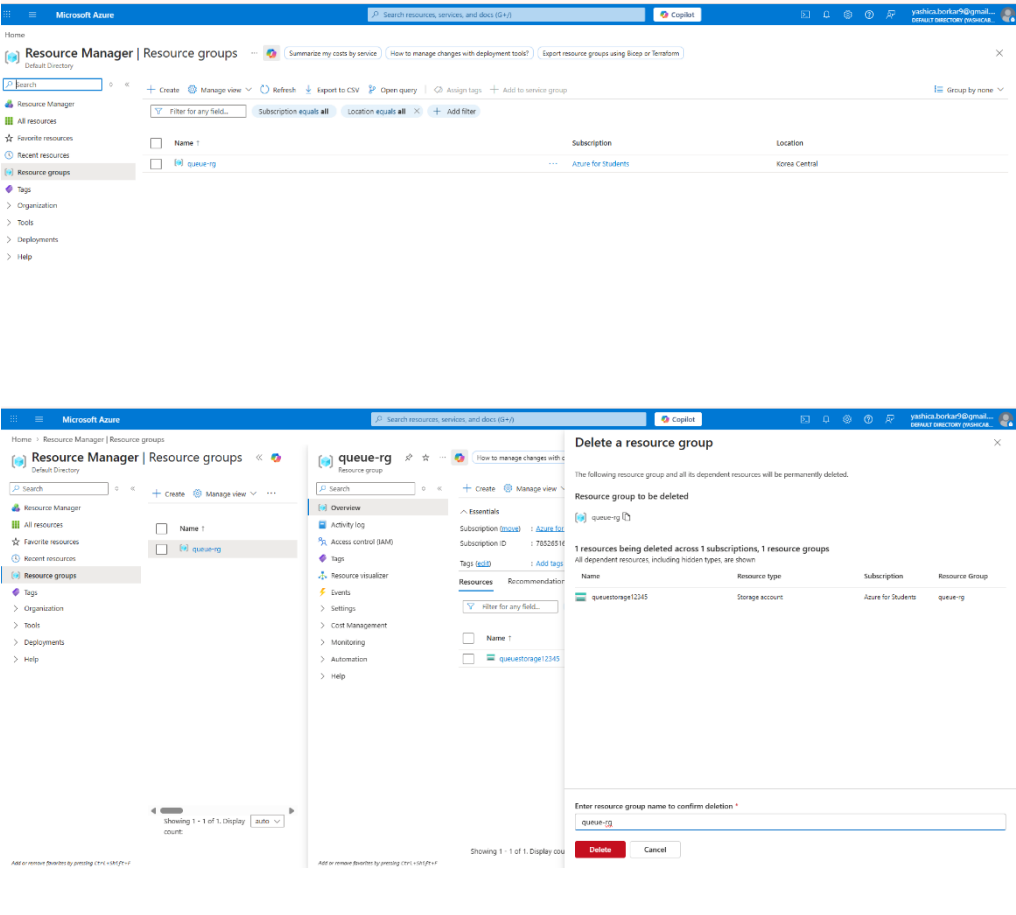
The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and a user profile. The main content area displays the 'Queues' section for the storage account 'queststorage12345'. The 'Queues' list shows several queues, including 'quickstartqueues-330b050-19af-4ab4-aeff-a0952a09033c' and 'quickstartqueues-30d34f25-183e-4149-a547-20ac26f27afa'. The 'quickstartqueues-30d34f25-183e-4149-a547-20ac26f27afa' queue is highlighted, showing its details and a list of messages.

Queue	URI
☐ quickstartqueues-330b050-19af-4ab4-aeff-a0952a09033c	https://queststorage12345.queue.windows.net/quickstartqueues-330b050-19af-4ab4-aeff-a0952a09033c
☐ quickstartqueues-56228416-5627-470f-ba2b-a87b782a476	https://queststorage12345.queue.windows.net/quickstartqueues-56228416-5627-470f-ba2b-a87b782a476
☐ quickstartqueues-1c3a2777-ba18-4acc-ba15-ba1509f87df	https://queststorage12345.queue.windows.net/quickstartqueues-1c3a2777-ba18-4acc-ba15-ba1509f87df
☐ quickstartqueues-a0b142f0-2aef-4096-623a-15e66a2d971f	https://queststorage12345.queue.windows.net/quickstartqueues-a0b142f0-2aef-4096-623a-15e66a2d971f
☐ quickstartqueues-cdf723f1-1d9d-412e-a8ec-c81300409d93	https://queststorage12345.queue.windows.net/quickstartqueues-cdf723f1-1d9d-412e-a8ec-c81300409d93
☐ quickstartqueues-d496a21b-c0bf-404d-b1be-a2d844320b12	https://queststorage12345.queue.windows.net/quickstartqueues-d496a21b-c0bf-404d-b1be-a2d844320b12

Step 11: Clean Up Resources

To avoid unnecessary charges:

- 1. Go to Resource Groups in Azure
- 2. Select queue-rg
- 3. Click Delete resource group



Result

Successfully created a queue, added messages, updated messages, retrieved messages, and deleted the queue using Azure Queue Storage and Python.

Conclusion

Learnings / Takeaways

- Understood the concept of asynchronous communication using Azure Queue Storage.
- Learned how to securely connect to Azure services using connection strings and environment variables.
- Gained practical experience in performing queue operations:
 - Creating a queue
 - Sending (adding) messages
 - Peeking messages without removing them
 - Updating message content
 - Receiving and deleting messages
- Understood the concept of decoupling, where producer and consumer work independently.

	• Learned that messages remain in the queue until processed and deleted, ensuring reliability. • Observed how queues improve scalability and fault tolerance of applications. • Understood the importance of resource cleanup to avoid unnecessary costs.
	Applications / Use Cases • Background Task Processing: Handling tasks like email sending, report generation, and image processing. • Microservices Communication: Enabling communication between independent services. • Order Processing Systems: Managing and processing orders sequentially in applications. • Event-Driven Systems: Triggering and processing events asynchronously. • Load Leveling: Managing high traffic by processing requests at a steady rate. • Data Processing Pipelines: Handling multi-stage data workflows efficiently. • Distributed Systems: Ensuring reliable communication between different system components. • Retry Mechanisms: Reprocessing failed tasks to improve system robustness.
References	https://learn.microsoft.com/en-us/azure/storage/queues/storage-queues-introduction Quickstart: Azure Queue Storage client library for Python - Azure Storage Microsoft Learn https://portal.azure.com/